How the OOD score gates the conditional counterfactual

Abstract

The attribution score and the OOD score answer different questions. Random-Fourier MMD^2, and the squared shift read off that kernel, say whether today's purchase law moved. The OOD score says whether the counterfactual conditioner was inside the training support of the diffusion model. A cell that fails the score does not get to speak as a headline effect. This note writes that gate down as three numbers, one action, and an eight-row evaluation. It is not a sixth frozen layer, and it does not retouch the five-layer calls from the landing case.

1 What the score is allowed to do

The conditional DDPM estimates $p(Y \mid x)$ on the x it was trained on. The counterfactual replaces the spiked IP block and rebuilds the lag vector. That new x^{cf} can leave the training marginal even when the intervention itself is the one the query asked for. Scoring Y there measures the model's extrapolation, not the conditional law the query named.

So the OOD score sits in front of the attribution, not inside it.

Object	Question	Allowed to change
Δ^{cf}, δ^2	Did the purchase law move, and by how much?	The reported effect
Support score s_∞	Is this conditioner inside the training box?	Whether the cell enters the headline
Localizer FSDS_j	Which coordinate pushed it out?	Clamp that coordinate, or drop the cell

The OOD score is not an effect size. A large FSDS_j does not mean cohort j drove purchases. It means coordinate j is why this question left the support.

2 Three numbers

The conditioner is the length-21 vector used by the denoiser: 14 IP lags, 3 covariates, overlap, and the 3 distractor blocks at lag 10. The reference cloud is the training design matrix, with coordinate mean μ and coordinate scale σ.

Support.

$$s_\infty(x) = \max_{j \le 21} \left| \frac{x_j - \mu_j}{\sigma_j} \right|.$$

The cell is in support when $s_\infty(x) \le 4$. The threshold is a coordinate box, not a joint density. A vector that is typical on every margin and unusual as a combination can pass. That is a stated limit of this score, not a hidden one.

Severity. On the standardized cloud, let $d_5(x)$ be the Euclidean distance from x to its 5th nearest training neighbor. This is the number a clamp has to reduce. It is not compared with 4. The box score and the neighbor score are two different geometries.

Driver.

$$\mathrm{FSDS}_j(x) = d_5(x) - d_5(x^{(j)}),$$

where $x^{(j)}$ replaces coordinate j by the cloud mean and leaves the rest alone. The largest coordinate is the one the fallback is allowed to touch. On a one-coordinate spike the identity $d_5(x) - d_5(x^{(j_\star)}) = \mathrm{FSDS}_{j_\star}$ is exact, and the run uses that identity as a check.

3 The chain, in order

The score empowers the attribution only by walking this list. Skipping a step is how an extrapolation gets quoted as an effect.

1. Build x^{cf} by putting the spike day back on the reference path. Later innovations, covariates, and distractors stay at their observed values.
2. Score $s_\infty(x^{\mathrm{cf}})$ against the training marginal. Do this before any MMD.
3. If $s_\infty \le 4$, the cell may enter Δ^{cf} and δ^2. The OOD score has nothing more to say.
4. If $s_\infty > 4$, compute FSDS_j on the standardized cloud.
5. One coordinate carries the distance: clamp that coordinate to the training mean, rescore, and label the cell clamped. The clamped value is a fallback, not the headline.
6. Several coordinates share the distance: drop the cell. Do not clamp a handful of them and keep the MMD.
7. The headline window is eligible only if its own conditioner passed step 3. A high support rate on other cells does not vouch for it.
8. δ^2 stays the size that gets quoted. s_∞ and FSDS_j never replace it.

4 Who is allowed into the headline

Passing $s_\infty(x^{\mathrm{cf}}) \le 4$ is not the headline decision. The headline is one sentence: on this window, putting the IP block back on the reference path lowers today's purchases. That sentence uses both batches, so the conditioner has to clear the edit and the background separately. A coordinate the replacement does not touch is background. On a lag-10 query the replacement touches one coordinate, b_0 at lag 10.

1. Name the edit set $J = \{j : x_j^{\mathrm{cf}} \neq x_j^{\mathrm{obs}}\}$. Everything else is background and is held fixed on purpose.
2. The edit is inside after the replacement: $\max_{j \in J} |z_j^{\mathrm{cf}}| \le 4$. If this fails, the reference path itself left the training box. The cell does not enter the headline.

3. The edit was not already an excuse to extrapolate: report $\max_{j \in J} |z_j^{\text{obs}}|$ next to it. Entering is the case where the spike coordinate moves toward the training center, not away from it.

4. Background is inside on the shared coordinates: $\max_{j \notin J} |z_j| \leq 4$. The same z appears on x^{obs} and x^{cf}, because those coordinates were not edited. If this fails, the observed batch is a real outcome at an unusual covariate and the counterfactual batch is a model draw at that same unusual covariate. The gap mixes the IP edit with extrapolation error. The cell is refused, and it is labeled background OOD, not "the counterfactual left the support."

5. Do not clamp a background coordinate and then call the result the headline. Clamping x or a distractor changes the facts the query said to hold fixed. A clamp is legal only for an edit coordinate that step 2 failed, and the clamped number is a fallback.

6. The two-batch sign has to match the edit. Here the spike is a positive dose and the kernel weight is positive, so the headline requires $\overline{Y}_{\text{obs}} - \overline{Y}_{\text{cf}} > 0$. A large MMD with the opposite sign, or with a near-zero mean gap, does not enter. The distance is not the direction.

7. The factual pair has to stay small. Δ^{cf} is the observed batch against the reference-path batch. Δ^{base} is the observed batch against a model batch that still carries the spike. The headline requires Δ^{cf} well above Δ^{base}, so the gap is the edit and not the sampler.

The dose-1 window is episode 42, site 0, horizon 10. J is only b_0 at lag 10. That coordinate moves from raw 1.063 to 0.063, and from $z = +2.53$ to $z = +0.09$. The box score of the whole vector stays $s_\infty = 2.58$, and the maximizing coordinate is the demographic covariate, which the edit did not touch. Signed mean gap +1.009. $\Delta^{\text{cf}} = 0.674$ against $\Delta^{\text{base}} = 0.021$. Steps 2 through 7 all pass, so this conditioner enters the headline. The sentence that enters is the signed gap, not s_∞.

The two cells with $s_\infty > 4$ fail step 4, not step 2. Episode 47, site 5, horizon 6: the demographic covariate is at $z = 4.14$ on both the observed and the reference conditioner. Episode 43, site 7, horizon 10: distractor b_1 at lag 10 is at $z = 4.18$ on both. The IP replacement did not move either coordinate. They are refused because the background is outside the box. They are not examples of the counterfactual path leaving the support, and they are not clamped.

5 What this run actually did

The validation grid is 10 sites, 16 horizons, 12 windows: 1920 conditioners. 1918 passed $s_\infty \leq 4$, which is a rate of 0.999. Both refusals are background OOD, computed above. Site 7 at lag 10 is the distractor cell, 11 of 12 windows inside the box. Its raw score is 0.019, and the missed window is not the headline. The dose-1 window entered under the seven-step rule, not under the box score alone.

The dose-response construction is the one place the code drops out-of-support rows before a concordance is computed. The site-mean Δ^{cf} still averages every window, including the one site-7 miss. On this panel that is one cell. The declaration is stricter than that average: step 7 above says the headline cell must pass on its own, and a site mean that will be read as an effect should use the same mask. That mask is not yet applied to the site table.

The localizer was not run on those 1920 real conditioners. It was run on a planted spike: lag-10 distractor b_2, coordinate 19, set to 4.5 training standard deviations, with every other standardized coordinate at 0. FSDS named b_2 at lag 10, score 1.42, and the next two coordinates scored 0. Neighbor distance fell from 4.05 to 2.63 after clamping that coordinate. The difference is 1.42, which

matches the score. The residual 2.63 is the 5th-neighbor distance of the cloud mean; clamping does not make a point identical to a training row. The real attribution path never entered this branch.

6 Evaluation list

Each row is a check on the gate. "This run" is the linear ten-site validation, not a new experiment. A fail here does not rewrite L1–L5. L5 still fails on the raw ratio, and L3 still fails on Wasserstein.

	Check	Pass condition	This run	Call
1	Coverage	Share of real x^{cf} with $s_\infty \le 4$ is reported, and the headline cell is among them	1918/1920; headline in support	pass
2	Gate binds	Out-of-support cells are counted, and the dose construction ignores them	2 cells out; dose uses the mask	pass
3	Site means	Any Δ that is read as an effect uses the same mask	Site table still averages the site-7 miss	gap
4	Planted driver	Argmax of FSDS is the planted coordinate	b_2 at lag 10, score 1.42	pass
5	Clamp identity	d_5 falls by $\mathrm{FSDS}_{j_\star}$ and by nothing else	$4.05 - 2.63 = 1.42$	pass
6	Real-path localizer	FSDS is computed on real out-of-support cells, not only on a plant	Not run; the two real misses were not localized	gap
7	Separation	δ^2 is the quoted size; s_∞ is not	δ^2 grid $0.808, 0.798, 0.787$; oracle mean of squares 0.754	pass
8	Joint hole	State that s_∞ is a box. A joint OOD point with small margins is a known miss	Stated. Not probed on this panel	open

Rows 3 and 6 are the two places the written chain is ahead of the code. Neither one changes the driver on this panel: the missed cell is a negative control, and the headline did not need a clamp. They are still listed, because a later panel with a real excursion would otherwise quote an extrapolated MMD as a site effect and would have no coordinate to clamp.

7 What stays put

The five-layer card is unchanged. Primary driver: site 0, lag 10, IP block, weekly scale. Headline $\Delta^{\mathrm{cf}} = 0.674$ with interval $[0.442, 0.906]$. Squared shift across the bandwidth grid: coefficient of variation 0.01. Raw ratio across that grid: 8.9 to 46.4, and that clause still fails. Feature-only Wasserstein floor: 0.166, model 0.245, and the 0.10 line still sits under the floor.